Quick Work

Page No.:

Date:

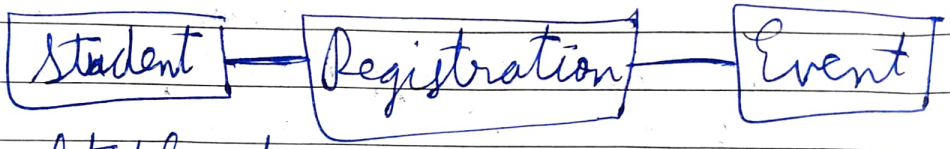
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Relationship

1. Generalization
2. Association: It indicates that two classes are related or they interact with each other.

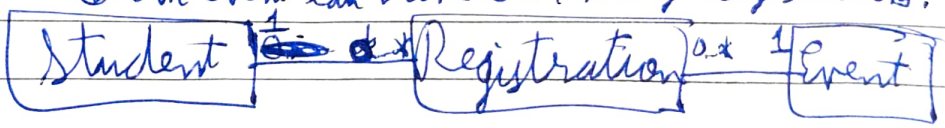
For eg: (a) A student makes a registration
(b) A registration is associated with an event



Multiplicity

Notation	Meaning
1	One
0..1	0 or 1
*	many
0..*	0 or many
1..*	1 or many.

For eg: (a) A student can have 0 or more registration.
(b) An event can have 0 or many registrations.



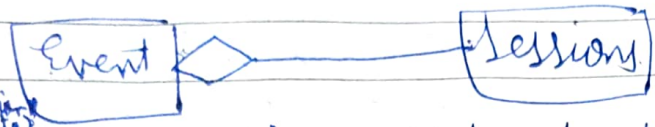
Quick Work

Page No.:

Date:

M T W T F S S

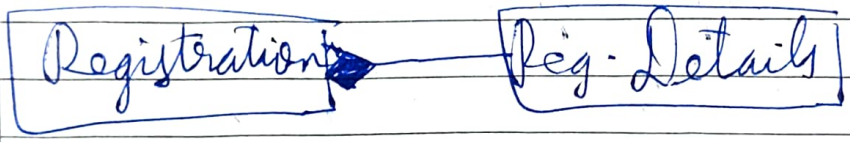
(3) Aggregation : ⇒ suppose an event can have multiple sessions.



why aggregation not multiplicity?

⇒ An event is composed of sessions, but a session can conceptually exist independently of a particular event.

(4) Composition ⇒ suppose registeration contains registration details & those details have no meaningful independent existence outside that registration.



- Collaboration diagram

⇒ It shows the objects involved in a scenario, their relationships, the messages exchanged b/w them & the order in which those messages occur..

Quick Work

Page No.:

Date:

M T W T F S S

Student 1 : Student

↓ 1 : register()

System : CampusHub

↓ 2 : checkSeat()

Event 1 : Event

↓ 3 : seatAvailable()

System : CampusHub

↓ 4 : CreateReg()

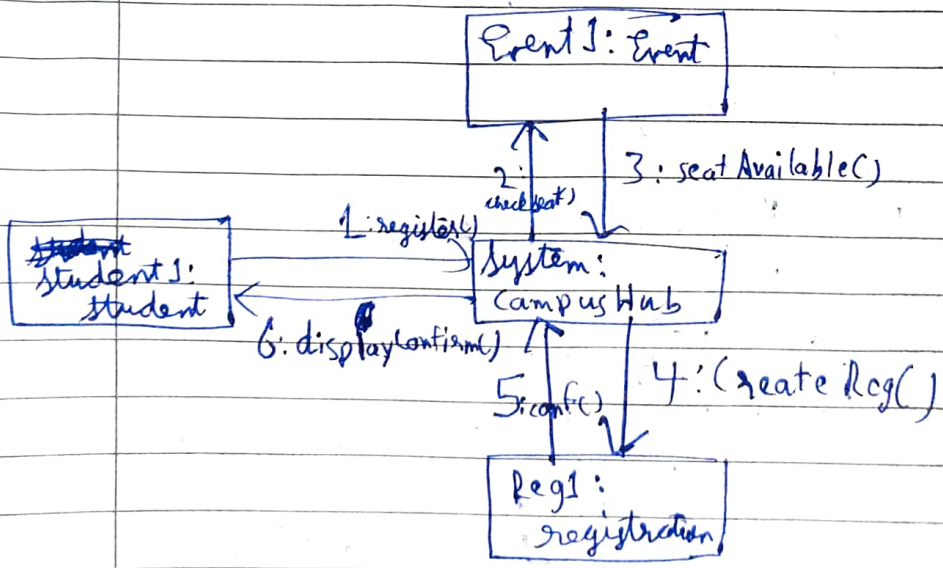
Reg 1 : Registration

↓ 5 : Conf()

System : CampusHub

↓ 6 : displayConfirm()

Student 1 : Student



Quick Work

Page No.:

Date:

M T W T F S S

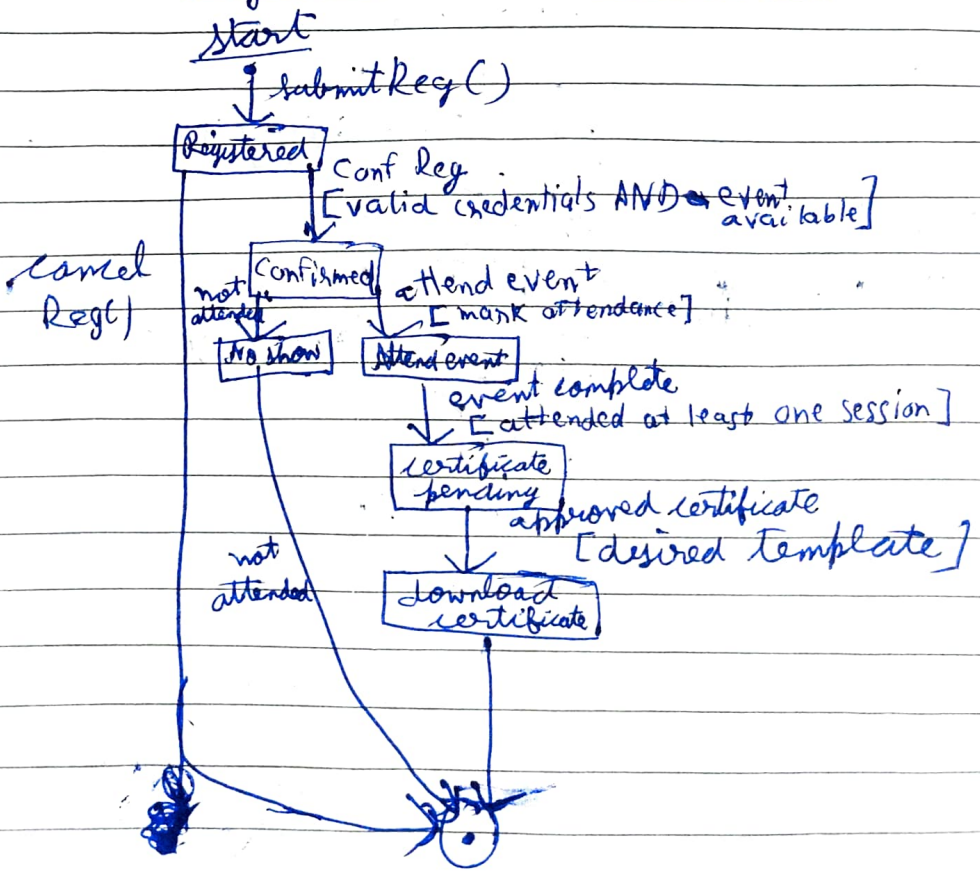
State Diagram

→ A state represents the condition or situation of an object at a particular point in time.

→ A state transition requires an event or trigger that moves an object from one state to another.

→ Suppose an event can be published only if it has been approved.

→ In this case, the transition occurs only when the cond^n is satisfied. Such a condition is called guard condition → in square bracket



Quick Work

Page No.:

Date:

M T W T F S S

- Component diagram

→ A component is a modular & replacable part of a computer system that provides specific functionality.

→ It is needed because a large system becomes difficult to develop, modify, maintain, test & difficult for multiple developers to work on.